Advance Analytics

1) The Sales Manager wants a list of all customers from Australia whose birthdate is available for a regional marketing campaign.

```
USE datawarehouse;

SELECT
    customer_number,
    first_name,
    last_name,
    country,
    gender,
    birthdate
FROM gold.dim_customers
WHERE country = 'Australia'
    AND birthdate IS NOT NULL
ORDER BY
    last_name ASC,
    first_name ASC;
```

2) Marketing wants to target married female customers from Australia and the United States who were born between 1980 and 1999.

```
USE datawarehouse;

SELECT
    customer_number,
    first_name,
    last_name,
    country,
    gender,
    marital_status,
    birthdate
FROM gold.dim_customers
WHERE
    gender = 'Female'
    AND country IN ('Australia', 'United States')
    AND marital_status = 'Married'
    AND birthdate BETWEEN '1980-01-01' AND '1999-12-31'
    AND birthdate IS NOT NULL
ORDER BY
    birthdate DESC,
    last_name ASC;
```

3) Management wants to identify the top five countries generating the highest revenue.

```
USE datawarehouse;

SELECT TOP 5
    d.country,
    COUNT(f.order_number) AS total_orders_by_country,
    SUM(f.sales) AS total_sales_by_country
FROM gold.fact_sales AS f
INNER JOIN gold.dim_customers AS d
    ON f.customer_id = d.customer_id
GROUP BY d.country
ORDER BY total_sales_by_country DESC;
```

4) Find customers who have never placed an order.

```
USE datawarehouse;

SELECT
    d.customer_number,
    d.first_name,
    d.last_name,
    d.country
FROM gold.dim_customers AS d
LEFT JOIN gold.fact_sales AS f
    ON f.customer_id = d.customer_id
WHERE f.customer_id IS NULL
ORDER BY
    d.country,
    d.last_name,
    d.first_name;
```

5) Find the highest-selling product in every country.

```
SELECT *
FROM
(
    SELECT
        RANK() OVER
        (
            PARTITION BY t.country
            ORDER BY t.total_sales DESC
        ) AS rnk,
        *
    FROM
    (
        SELECT
            dp.product_name,
            dc.country,
            SUM(f.sales) AS total_sales
        FROM gold.fact_sales AS f
        INNER JOIN gold.dim_customers AS dc
            ON f.customer_id = dc.customer_id
        INNER JOIN gold.dim_products AS dp
            ON f.product_key = dp.product_key
        GROUP BY
            dc.country,
            dp.product_name
    ) t
) tt
WHERE tt.rnk = 1;
```

6) The Finance team wants to monitor cumulative monthly sales over time.

```
SELECT
    t.sales_year,
    t.sales_month_,
    t.total_sales,
    SUM(t.total_sales)
    OVER
    (
        ORDER BY
            t.sales_year,
            t.sales_month
        ROWS BETWEEN UNBOUNDED PRECEDING
        AND CURRENT ROW
    ) AS running_total
FROM
(
    SELECT
        YEAR(order_date) AS sales_year,
        DATENAME(MONTH, order_date) AS sales_month_,
        MONTH(order_date) AS sales_month,
        SUM(sales) AS total_sales
    FROM gold.fact_sales
    GROUP BY
        YEAR(order_date),
        MONTH(order_date),
        DATENAME(MONTH, order_date)
) t;
```